

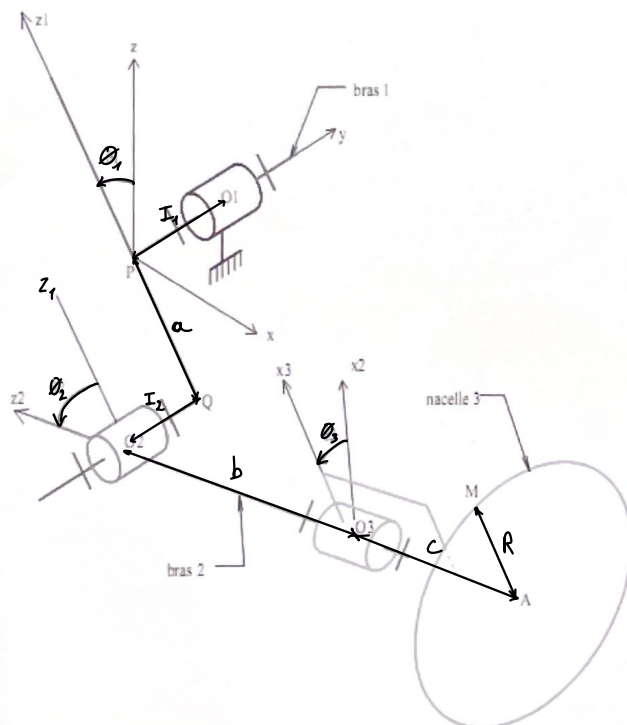
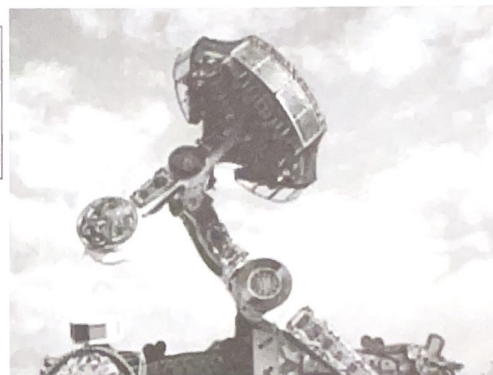
EXERCICE D'APPLICATION : MAGIC ARMS

Compétences :

- ☐ Déterminer les relations entre les grandeurs cinématiques.
- ☐ Caractériser le mouvement d'un repère par rapport à un autre repère

Le Magic Arms est un manège fabriqué par la société WAAGNER-BIRO. Ses mouvements simultanés autour de trois axes, désorientent les 39 passagers embarqués qui ne savent plus reconnaître le dessus du dessous pendant quelques minutes.

La modélisation adoptée est la suivante :



On pose:

$$PO_1 = l_1$$

$$QP = a$$

$$O_2Q = l_2$$

$$O_3O_2 = b$$

$$AO_3 = c$$

$$AM = R$$

$$\theta_1 = (\vec{z}, \vec{z}_1) = (\vec{x}, \vec{x}_1)$$

$$\theta_2 = (\vec{z}_1, \vec{z}_2) = (\vec{x}_1, \vec{x}_2)$$

$$\theta_3 = (\vec{x}_2, \vec{x}_3) = (\vec{y}_2, \vec{y}_3)$$

Question 1. Effectuer le graphe des liaisons et les figures de projections montrant les angles $\theta_1; \theta_2; \theta_3$.

Question 2. Déterminer les torseurs suivants : $\{V_{1/0}\}, \{V_{2/1}\}, \{V_{3/2}\}$

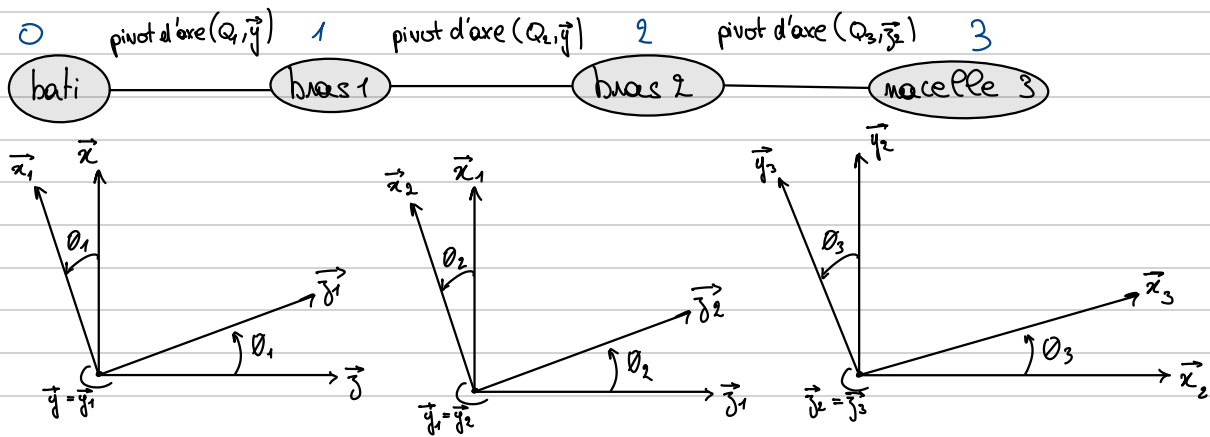
Question 3. Calculer $\overline{V(M \in 3/0)}$ en déterminant le torseur $\{V_{3/0}\}$ au point M.

Question 4. Calculer $\overline{V(M \in 3/0)}$ par dérivation et retrouver le résultat.

Question 5. On suppose dans la suite de l'exercice que $\theta_3 = 0$. Calculer $\overline{\Gamma(M \in 3/0)}$

$$\text{Solution : } \overline{V(M, 3/0)} = R\dot{\theta}_3\vec{y}_3 + (\dot{\theta}_1 + \dot{\theta}_2)[-(b+c)\vec{x}_2 - R\cos\theta_3\vec{z}_2] - a\dot{\theta}_1\vec{x}_1$$

Question 1:



Question 2:

Entre le bâti et le bras 1: $\{V_{1/0}\} = \begin{Bmatrix} \dot{\theta}_1 \vec{y}_1 \\ \vec{0} \end{Bmatrix}_{O_1}$

Entre le bras 1 et le bras 2: $\{V_{2/1}\} = \begin{Bmatrix} \dot{\theta}_2 \vec{y}_1 \\ \vec{0} \end{Bmatrix}_{O_1}$

Entre le bras 2 et le morceau 3: $\{V_{3/2}\} = \begin{Bmatrix} \dot{\theta}_3 \vec{y}_2 \\ \vec{0} \end{Bmatrix}_{O_2}$

Bonus: Trouver la vitesse du point M par dérivation par rapport au bâti: $\vec{V}(M/R_0)$

$$\begin{aligned} \vec{O_1 M} &= \vec{O_1 P} + \vec{P Q} + \vec{Q O_2} + \vec{O_2 O_3} + \vec{O_3 A} + \vec{A M} \\ &= -I_1 \vec{y}_1 - a \vec{y}_1 - I_2 \vec{y}_1 - b \vec{y}_2 - c \vec{y}_2 + R \vec{x}_3 \end{aligned}$$

$$\vec{V}(M/R_0) = \frac{d}{dt}(\vec{O_1 M})_{R_0}$$

$$= \frac{d}{dt}(-I_1 \vec{y}_1 - a \vec{y}_1 - I_2 \vec{y}_1 - b \vec{y}_2 - c \vec{y}_2 + R \vec{x}_3)_{R_0}$$

$$= \frac{d}{dt}(-I_1 \vec{y}_1)_{R_0} + \frac{d}{dt}(-a \vec{y}_1)_{R_0} + \frac{d}{dt}(-I_2 \vec{y}_1)_{R_0} + \frac{d}{dt}(-b \vec{y}_2)_{R_0} + \frac{d}{dt}(-c \vec{y}_2)_{R_0} + \frac{d}{dt}(R \vec{x}_3)_{R_0}$$

$$= -I_1 \frac{d\vec{y}_1}{dt}_{R_0} - a \frac{d\vec{y}_1}{dt}_{R_0} - I_2 \frac{d\vec{y}_1}{dt}_{R_0} - b \frac{d\vec{y}_2}{dt}_{R_0} - c \frac{d\vec{y}_2}{dt}_{R_0} + R \frac{d\vec{x}_3}{dt}_{R_0}$$

$$= R \frac{d\vec{x}_3}{dt}_{R_0} - a \frac{d\vec{y}_1}{dt}_{R_0} - b \frac{d\vec{y}_2}{dt}_{R_0} - c \frac{d\vec{y}_2}{dt}_{R_0}$$

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Changement de repère

$$\frac{d\vec{x}_3}{dt}/R_0 = \frac{d\vec{x}_3}{dt}/R_3 + \vec{\Omega}(3/0) \wedge \vec{x}_3$$

$$= \vec{\Omega}(3/0) \wedge \vec{x}_3 \Rightarrow \vec{\Omega}(3/0) = \vec{\Omega}(3/2) + \vec{\Omega}(2/1) + \vec{\Omega}(1/0) = \dot{\theta}_3 \vec{z}_2 + \dot{\theta}_2 \vec{y}_1 + \dot{\theta}_1 \vec{y}$$

$$= (\dot{\theta}_3 \vec{z}_2 + \dot{\theta}_2 \vec{y}_1 + \dot{\theta}_1 \vec{y}) \wedge \vec{x}_3$$

$$= \dot{\theta}_3 (\vec{z}_2 \wedge \vec{x}_3) + \dot{\theta}_2 (\vec{y}_1 \wedge \vec{x}_3) + \dot{\theta}_1 (\vec{y} \wedge \vec{x}_3) \Rightarrow \vec{z}_2 \wedge \vec{x}_3 = \vec{z}_3 \wedge \vec{x}_3 = \vec{y}_3$$

$$\frac{d\vec{x}_3}{dt}/R_0 = \dot{\theta}_3 \vec{y}_3 - \dot{\theta}_2 \cos(\theta_3) \vec{z}_3 - \dot{\theta}_1 \cos(\theta_3) \vec{z}_3$$

$$\vec{y}_1 \wedge \vec{x}_3 = \vec{y}_2 \wedge \vec{x}_3 = -\sin(\frac{\pi}{2} - \theta_3) \vec{z}_3 = -\cos(\theta_3) \vec{z}_3$$

$$\vec{y} \wedge \vec{x}_3 = \vec{y}_2 \wedge \vec{x}_3 = -\cos(\theta_3) \vec{z}_3$$

$$\frac{d\vec{y}_1}{dt}/R_0 = \frac{d\vec{y}_1}{dt}/R_1 + \vec{\Omega}(1/0) \wedge \vec{y}_1$$

$$= \vec{\Omega}(1/0) \wedge \vec{y}_1 \Rightarrow \vec{\Omega}(1/0) = \dot{\theta}_1 \vec{y}$$

$$= \dot{\theta}_1 \vec{y} \wedge \vec{y}_1$$

$$= \dot{\theta}_1 (\vec{y} \wedge \vec{y}_1) \Rightarrow \vec{y} \wedge \vec{y}_1 = \vec{y}_1 \wedge \vec{y}_1 = \vec{x}_1$$

$$\frac{d\vec{y}_1}{dt}/R_0 = \dot{\theta}_1 \vec{x}_1$$

$$\frac{d\vec{y}_2}{dt}/R_0 = \frac{d\vec{y}_2}{dt}/R_2 + \vec{\Omega}(2/0) \wedge \vec{y}_2$$

$$= \vec{\Omega}(2/0) \wedge \vec{y}_2 \Rightarrow \vec{\Omega}(2/0) = \vec{\Omega}(2/1) + \vec{\Omega}(1/0) = \dot{\theta}_2 \vec{y}_1 + \dot{\theta}_1 \vec{y}$$

$$= (\dot{\theta}_2 \vec{y}_1 + \dot{\theta}_1 \vec{y}) \wedge \vec{y}_2$$

$$= \dot{\theta}_2 (\vec{y}_1 \wedge \vec{y}_2) + \dot{\theta}_1 (\vec{y} \wedge \vec{y}_2) \Rightarrow \vec{y}_1 \wedge \vec{y}_2 = \vec{y}_2 \wedge \vec{y}_2 = \vec{x}_2$$

$$\frac{d\vec{y}_2}{dt}/R_0 = \dot{\theta}_2 \vec{x}_2 + \dot{\theta}_1 \vec{x}_2$$

$$\vec{y} \wedge \vec{y}_2 = \vec{y}_2 \wedge \vec{y}_2 = \vec{x}_2$$

$$V(M/R_0) = R \left(\dot{\theta}_3 \vec{y}_3 - \dot{\theta}_2 \cos(\theta_3) \vec{z}_3 - \dot{\theta}_1 \cos(\theta_3) \vec{z}_3 \right) - a \left(\dot{\theta}_1 \vec{x}_1 \right) - b \left(\dot{\theta}_2 \vec{x}_2 + \dot{\theta}_1 \vec{x}_2 \right) - c \left(\dot{\theta}_2 \vec{x}_2 + \dot{\theta}_1 \vec{x}_2 \right)$$

$$\text{Solution : } V(M, 3/0) = R \dot{\theta}_3 \vec{y}_3 + (\dot{\theta}_1 + \dot{\theta}_2) [-(b+c) \vec{x}_2 - R \cos \theta_3 \vec{z}_2] - a \dot{\theta}_1 \vec{x}_1$$

Question 3:

$$\{V_{3/0}\} = \{V_{3/2}\} + \{V_{2/1}\} + \{V_{1/0}\}$$

$$\{V_{1/0}\} = \left\{ \frac{\dot{\vec{r}}_1}{\vec{0}} \right\}_p = \left\{ \begin{array}{c} \dot{\vec{r}}_1 \\ \dot{\vec{r}}_1 (-a\vec{x}_1 - (b+c)\vec{x}_2 - R\cos\theta_3\vec{y}_2) \end{array} \right\}_M$$

$$\{V_{2/1}\} = \left\{ \frac{\dot{\vec{r}}_2}{\vec{0}} \right\}_{o_2} = \left\{ \begin{array}{c} \dot{\vec{r}}_2 \\ \dot{\vec{r}}_2 (-(b+c)\vec{x}_2 - R\cos\theta_3\vec{y}_2) \end{array} \right\}_M$$

$$\{V_{3/2}\} = \left\{ \frac{\dot{\vec{r}}_3}{\vec{0}} \right\}_{o_3} = \left\{ \begin{array}{c} \dot{\vec{r}}_3 \\ R\dot{\theta}_3\vec{y}_3 \end{array} \right\}_M$$

$$\overline{V(M, 3/0)} = R\dot{\theta}_3\vec{y}_3 + (\dot{\theta}_1 + \dot{\theta}_2)(-(b+c)\vec{x}_2 - R\cos\theta_3\vec{y}_2) - \dot{\theta}_1 a\vec{x}_1$$